SCHOOL OF APPLIED SCIENCE AND OPEN LEARNING

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF CYBER SECURITY

CYS 335

ACCESS CONTROL AND INTRUSION DETECTION

SESSIONAL EXAMINATION

DATE: 04-Nov-2024 TIME: 14.00 - 14.00 HRS, 3 HOURS

Instructions to candidates

1. Write your **Name** and **Student ID** on the **Answer Sheet** provided
2. There are **Six (6) questions** in this paper. Attempt any **five (5).** Indicate on the Answer sheet your choice from the optional questions.
3. Write your Answers on the answer Sheet provided.
4. No Foreign material will be allowed in this examination.
5. The number of marks is shown in brackets **[ ]** at the end of each question or part question.
6. Clear handwriting and neat work is encouraged.
7. The total number of marks for this paper is **100**

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QUESTION 1

a) At times because of lack of expertise in certain areas, Organizations tend to contract some works from other organizations. You are a senior Cyber Security Specialist working closely with the contractor. How should temporary access be managed for a contractor working on a short-term project?

[3 Marks]

b) "Cyber Attacks are inevitable and we have to live with them". Therefore what role can honeypots play in enhancing an IDS's effectiveness? *[3 Marks]*

c) Explain how integrating access control with IDS contributes to a multi-layered defense strategy.

[3 Marks]

d) It has been argued that warning increase the sense of fear in potential criminals. In what ways can publicizing the presence of security cameras deter unauthorized access to sensitive areas?

[4 Marks]

e) What role does regular security training play in deterring employees from engaging in unsafe practices? [4 Marks]

f) What measures would you put in place to prevent data breaches in a cloud computing environment? [3 Marks]

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QUESTION 2

a) Tokens are said to have better security than passwords. Explain the mechanism of an asynchronous based token authentication [4 Marks]

b) How would you restore customer confidence in your organization following a publicized security incident? [2 Marks Each]

c) To easy implementation of access control, a number of concepts or abstract ideas have been developed. Describe the following implementation of an Access Control Matrix:
   a. Access Control List [3 Marks]
   b. Capability List [3 Marks]

d) There several password attacks that an attacker might use. You are required to identify Four(4) of them and suggest mitigation measure respectively. [8 Marks]

QUESTION 3

a) What steps should be taken to secure remote access to the corporate network? [5 Marks]

b) How would you use audit logs to support non-repudiation in case of a dispute? [5 Marks]

c) What steps should be taken if an IDS detects an unauthorized remote access attempt that bypassed the access control system? [5 Marks]

d) Explain how integrating access control with IDS contributes to a multi-layered defense strategy. [5 Marks]

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If access control logs indicate possible compromised credentials and an IDS confirms suspicious activity from the related account, how should the situation be handled? [5 Marks

QUESTION 4

Technical control is at the very core of access control. Technical controls called logical controls are the software tools used to restrict subject's access to objects.

Identify and explain four (4) of the control components [5 Marks Each

QUESTION 5

At a company named "NSHALIKO" the administration had a notion that every aspect of Cyber security is for the ICT Department of the organization and that administration had nothing to do with it. It was discovered that despite the ICT Department doing it's level best, there where still some security challenges here and there. You are now invited to present to the company missing gaps in cyber security to highlight to the management that it has a significant role to play in access control contrary to some ideas that this role is purely technical. Your presentation includes the following suggestions. Explain how each of them is important and may be used:

i) Security Awareness Training [4 Marks]

ii) Supervisory structure [4 Marks]

iii) Testing [4 Marks]

iv) Personel Controls [4 Marks]

v) Policies and procedures [4 Marks]

QUESTION 6

a) What role does regular security training play in deterring employees from engaging in unsafe practices? [3 Marks]

b) Auditing is an important part of system security. Elaborate briefly on what auditing constitutes. [3 Marks]

c) Mention and briefly explain 5 of the best practices of authentication and Access control [2 Marks Each]

d) How can a honeypot system be used to deflect attacks from critical systems? [4 Marks]

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